

Digital Thorium

Term Sheet

A Sovereign Production-Linked Claim on Clean Baseload Electricity · April 2026 · Abhilash Inumella & Anubhav Tiwari

INSTRUMENT		VARIANTS		RISK FRAMEWORK			
Name	Digital Thorium	Base token	Full float, no floor/cap	Construction risk (issued)	Zero (commissioning-gated)		
Type	Sovereign production-linked claim	Floored token	Floor at INR 4.0/kWh CPI-linked	Construction risk (warrant)	Borne by holder		
Closest comp.	Royalty trust / streamer	Capped token	Forgo upside above INR 12/kWh real	Operational risk	Cohort-pooled, 8% buffer		
Unit of account	1 kilowatt-hour-year (kWhy)	Construction warrant	Pre-commissioning, deep discount	Settlement risk	Statutory guarantee fund 5%		
Issuer	NTFC (statutory corporation)	Derivatives layer	Futures, options (year 2+)	Sovereign backstop	Statutory, not constitutional		
Statutory basis	National Thorium Finance Act, 2027	Retail leveraged	Prohibited first 5 years	Political reversal protect.	200M retail + Act notice rule		
Underlying	Thorium electricity, Indian fleet	TAX TREATMENT (PROPOSED FA 2027)					
Form	Exchange-listed fractional claim						
Tenor	Perpetual / 60-year asset life						
Settlement currency	INR	TDS (resident, indiv.)	Nil under INR 10K/yr, 10% above	Reinsurance	International, cohort outage		
ISSUANCE & PRICING		TDS (HUF, non-indiv.)	10% from rupee one	DOWNSIDE SCENARIOS (NAV IMPACT)			
		Income classification	Slab rate; concessional under review				
		LTCG (held >24 mo)	12.5%				
Launch price	INR 20 per kWhy	STCG	Slab rate	AI demand halves	-15% NAV trough, IRR to 9.6%		
Minimum ticket	INR 500	STT	0.05% sell side	PFBR slips 4 years	No haircut, opportunity cost only		
Issuance gating	Commissioned capacity only	NRI withholding	20%, DTAA reducible	Major cohort outage 6mo	-3 to -4% NAV, recovery 18mo		
Construction warrants	Separate class, 25–35% discount	Pension / insurance	Approved investment, concessional	Fusion shock (long-dated)	-25 to -40% LR curve, IRR to 6–7%		
Net pass-through	87.5%	FII status	Eligible, 10% cohort cap	Mitigation across all	Real cash flow continues		
Reference price	VWAP IEX + PXIL day-ahead	LIQUIDITY & COLLATERAL					
Smoothing	90-day moving average						
Methodology change	18 months notice + grandfather						
Distribution frequency	Quarterly	Year 1 bid-ask	≈200 bps, PD-quoted	AUDIT & GOVERNANCE			
First settlement	T+45 days post-issuance	Year 3 bid-ask	<50 bps target				
ECONOMICS (CENTRAL CASE)		Year 5 bid-ask	Listed bond comparable				
		Daily turnover target Y1	0.5% of float	Statutory audit	Comptroller & Auditor General		
		Daily turnover target Y3	1.5–2% of float	Tokenisation audit	Big Four firm		
Target real IRR	11.2%	Primary dealers	Panel of 8 at launch	Physical audit	AERB / IGCAR / WNA observer		
Target nominal IRR	14.5%	Retail buy-back window	Quarterly, NAV less 75 bps	Market conduct	SEBI		
Distribution yield (launch)	≈5.0%	LAS collateral (RBI)	Day-1, 35% haircut	Disclosure	Quarterly + real-time dashboard		
Capital appreciation target	6–8% real	IRDAI treatment	Approved Cat II, 1.25× charge	Parliamentary oversight	Annual report tabled		
Modified duration	≈12.6 years	SLR eligibility	Year 3+ subject to history	Governance committee	MoF + DAE + SEBI joint		
Macauley duration	≈14.0 years	LOCKUPS & OWNERSHIP CAPS					
Capacity factor (SS)	82%						
Real escalation assumption	1.5%						
Reference G-sec spread	≈750 bps	Retail tranche	No lockup	Holder rights	No ops voting; Tribunal redress		
Political risk premium	≈150 bps	Institutional anchor	12 months	PROGRAMME SIZING			
DEDUCTION STACK (% OF GROSS)		Foreign sovereign	24 months				
		NTFC strategic reserve	60 months				
		Foreign ownership cap	10% per cohort, aggregate				
Host state royalty	3.0%	Retail allocation target	50% of float	Pilot cohort (2027)	500 MW PFBR-derived		
Operator margin	8.0%	2030 target				5 GW tokenised	
NTFC management fee	1.0% (declines to 0.5%)	2035 target				20 GW tokenised	
Regulatory & audit	0.3%	2046 base case				50 GW commissioned	
Settlement & custody	0.1%	Steady-state stock				INR 7.2 lakh crore	
Net to holders	87.5%	Retail participation target				200 million households	
Statutory deduction cap	15% of gross	Annual fiscal benefit (SS)				INR 3.0–3.5 lakh crore	
						Oil import substitution	\$22 billion/yr at scale